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```
% HW1AE105A.m
% Used the mic cookbook.m as a template
% documentation and for set up [A] [B] and [C]: GettingStartedWithMICE.pdf
%
% March 30, 2019 stefano.campagnola@jpl.nasa.gov
%
% SET UP IS BELOW, THEN THE HOMEWORK PROBLEMS
% -----
```

[A] Set paths

- (1) Replace `path_to_generic_kernels` and `path_to_mice` with your path
- (2) Remember to use `'/'` for mac and linux, `'\'` for windows

```
path_to_generic_kernels = 'C:\Users\Owner\Downloads\Ae105_MICE
\generic_kernels';
path_to_mice             = 'C:\Users\Owner\Downloads\Ae105_MICE\mice';
addpath([path_to_mice, '\src\mice']);
addpath([path_to_mice, '\lib']);
```

[B] Load the datafiles (kernels)

- (1) leap-seconds

```
cspice_furnsh( [path_to_generic_kernels, '\lsk\naif0012.tls.pc']);
% (2) jovian satellite ephemerides
cspice_furnsh( [path_to_generic_kernels, '\spk\satellites\jup310.bsp']);
% (3) planets
cspice_furnsh( [path_to_generic_kernels, '\spk\planets\de421.bsp']);
% (4) gravity constants - you can also open this file with a text editor to
read its
% content
cspice_furnsh( [path_to_generic_kernels, '\pck\gm_de431.tpc']);
% (5) planetary constant - you can also open this file with a text editor to
read its
% content
cspice_furnsh( [path_to_generic_kernels, '\pck\pck00010.tpc']);
```

[C] Planetary Constant References

- (1) You can read details on `cspice_bodvcd` and on the `pck` file

(2) NOTE: Planets/Moons IDs are as follow

1 or 199	Mercury barycenter
2 or 299	Venus barycenter
3	Earth Moon Barycenter
301	Moon Barycenter
399	Earth Barycenter
4	Mars system barycenter
401	Phobos
402	Deimos
499	Mars
5	Jupiter system Baricenter

etc etc . Read [naif_ids.html](#) in the documentation for more info

```
% Clear the memory
%cspice_kclear
```

Problem 2: Elevation and Azimuth of Mars and Jupiter wrt surface of Earth

```
% Here is the necessary Setup
```

```
% Time and Dates
```

```
date0 = '2025 Apr 5 10:00:00 UTC'; % can edit this to any time desired
et0 = cspice_str2et(date0);
```

```
% Longitude and Latitude angles of Pasadena, CA on Earth
```

```
phi = deg2rad(34.1478);
theta = deg2rad(-118.1445);
```

```
% Rotation Vectors
```

```
U = [cos(phi)*cos(theta); cos(phi)*sin(theta); sin(phi)];
N = [-sin(phi)*cos(theta); -sin(phi)*sin(theta); cos(phi)];
E = [-sin(theta); cos(theta); 0];
```

```
rot_vec = [E,N,U]';
```

```
% Initialize the rotation matrix to align the earth with the ecliptic frame
```

```
Rot = cspice_pxform('ECLIPJ2000', 'IAU_EARTH', et0);
```

```
% Radius Correction (translates data from Earth's center to surface)
```

```
Radius_Correction = 6371 * U;
```

```
% Earth's state with respect to Mars (EwrtM) and with respect to Jupiter
```

```
% (EwrtJ) barycenters
```

```
EwrtM = cspice_spkpos('4', et0, 'ECLIPJ2000', 'NONE', '3');
EwrtJ = cspice_spkpos('5', et0, 'ECLIPJ2000', 'NONE', '3');
```

```
% Modified position states to be in the correct reference frame
```

```
% This is where EwrtM_ECEF is the EwrtM state in the ECEF (earth centered,
% earth fixed) reference frame and EwrtJ_ECEF is the EwrtJ state in the
% ECEF reference frame.
```

```

EwrtM_ECEF = Rot*EwrtM(1:3); % multiplying position coordinates of EwrtM by
    ECEF rotation matrix
EwrtJ_ECEF = Rot*EwrtJ(1:3); % multiplying position coordinates of EwrtJ by
    ECEF rotation matrix

% Topographically accurate EwrtM and EwrtJ by shifting the position states
% for EwrtM and EwrtJ from refering to the center of the Earth to its
% surface

EwrtM_topo = EwrtM_ECEF - Radius_Correction;
EwrtJ_topo = EwrtJ_ECEF - Radius_Correction;

% Create the directional vectors of EwrtM and EwrtJ
E_mars = dot(E,EwrtM_topo);
E_jupiter = dot(E,EwrtJ_topo);
N_mars = dot(N,EwrtM_topo);
N_jupiter = dot(N,EwrtJ_topo);
U_mars = dot(U,EwrtM_topo);
U_jupiter = dot(U,EwrtJ_topo);

% Convert the directional vectors of Earth wrt Mars and Jupiter to angles
% (azimuth and elevation)

% Convert directional vectors to angles function definition
Vect_2_ang = @(E,N,U,ewrt) deal(mod(atan2(E,N), 2*pi), atan2(U, norm(ewrt)));

% Applying the function above to obtain Azimuth and Elevation angles
[azimuth_M, elev_M] = Vect_2_ang(E_mars, N_mars, U_mars,EwrtM_topo);
[azimuth_J, elev_J] = Vect_2_ang(E_jupiter, N_jupiter, U_jupiter,EwrtJ_topo);

% Display the results

fprintf('Elevation of Mars wrt Earth')
rad2deg(elev_M)

fprintf('Elevation of Jupiter wrt Earth')
rad2deg(elev_J)

fprintf('Azimuth of Mar wrt Earth')
rad2deg(azimuth_M)

fprintf('Azimuth of Jupiter wrt Earth')
rad2deg(azimuth_J)

Elevation of Mars wrt Earth
ans =

    -0.6686

Elevation of Jupiter wrt Earth
ans =

   -24.1700

```

Azimuth of Mar wrt Earth

ans =

299.4779

Azimuth of Jupiter wrt Earth

ans =

329.0405

Problem 3

% Necessary Setup is needed

% Time and Dates

date3 = '2025 Jan 1 0:00:00 PDT';

et3 = cspice_str2et(date3);

et3_R = et0:3600:(et0+86400);% make a row vector of times for a day

et_Y = et3:86400:(et3+(3.15576*10^7)); % make a row vector of times for a
year

et_3Y = et3:86400:(et3+(9.46728*10^7)); % make a row vector of times for 3
years

date4 = '2025 Oct 1 0:00:00 PDT';

et4 = cspice_str2et(date4);

et_3M = et4:86400:(et4+(7.889399*10^6)); % make a row vector of times for 3
months

% Plot orbits of planets with respect to the sun propagating for a whole
% year (Jan 1st 2025 to Dec 31st 2027)

% planets array

planets = ['1' '2' '3' '4' '5' '6' '7' '8'];

figure;

hold on;

xlabel('km')

ylabel('km')

title("Orbits of Planets around the Sun in Space Jan 1st 2025 - Dec 31st
2025")

for j = 1:8

pos = zeros(3, 8);

for k = 1:numel(et_Y)

pos(:, k) =

cspice_spkpos(planets(j), et_Y(k), 'ECLIPJ2000', 'NONE', '0');

end

plot (pos(1,:), pos(2,:), '-');

end

legend('Mercury', 'Venus', 'Earth', 'Mars', 'Jupiter', 'Saturn', 'Uranus', 'Neptune');

hold off;

```

% Jan 1st 2025 to December 31st 2027 Mars wrt Sun radius/velocity plot

figure;
% Initialize position and velocity states for Mars wrt Sun
r_m = zeros(1, numel(et_3Y));
v_m = zeros(1, numel(et_3Y));
for k = 1:numel(et_3Y)
    MwrtS = cspice_spkezr('499', et_3Y(k), 'ECLIPJ2000', 'NONE', '0');
    r = MwrtS(1:3);
    v = MwrtS(4:6);
    r_m(k) = norm(r);
    v_m(k) = norm(v);
end
figure;
plot(et_3Y, r_m, '-');
xlabel('km')
ylabel('km')
title("Orbital radius Mars around the Sun in Space Jan 1st 2025 - Dec 31st 2027")

figure;
plot(et_3Y, v_m, '-');
xlabel('km')
ylabel('km')
title("Orbital velocity Mars around the Sun in Space Jan 1st 2025 - Dec 31st 2027")

% % Oct 1st 2025 to December 31st 2025 Moon wrt Earth (mwrtE) radius and speed
% plots
% same approach as plot right above (MwrtS)

figure;
r_e = zeros(1, numel(et_3M));
v_e = zeros(1, numel(et_3M));
for k = 1:numel(et_3M)
    mwrtE = cspice_spkezr('3', et_3M(k), 'ECLIPJ2000', 'NONE', '301');
    r = mwrtE(1:3);
    v = mwrtE(4:6);
    r_e(k) = norm(r);
    v_e(k) = norm(v);
end

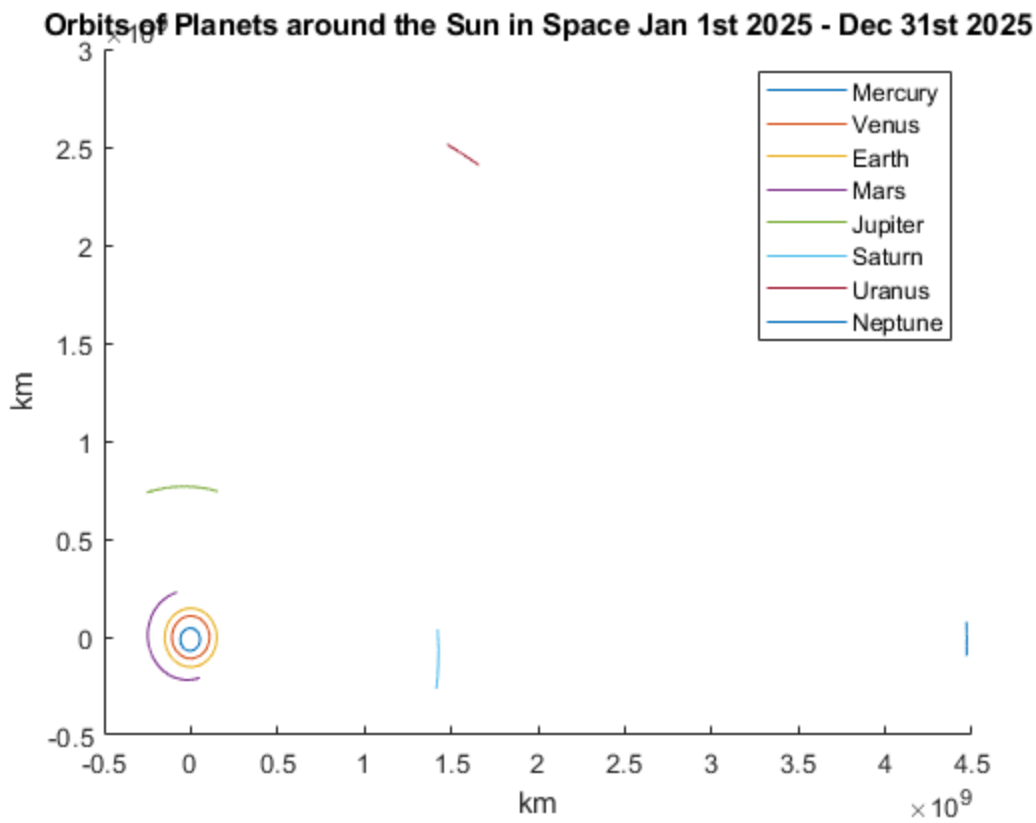
plot(et_3M, r_e, '-');
xlabel('km')
ylabel('km')
title("Orbital radius Moon wrt Earth in Space Oct 1st 2025 - Dec 31st 2025")

plot(et_3M, v_e, '-');
xlabel('km')
ylabel('km')
title("Orbital velocity Moon wrt Earth in Space Oct 1st 2025 - Dec 31st 2025")

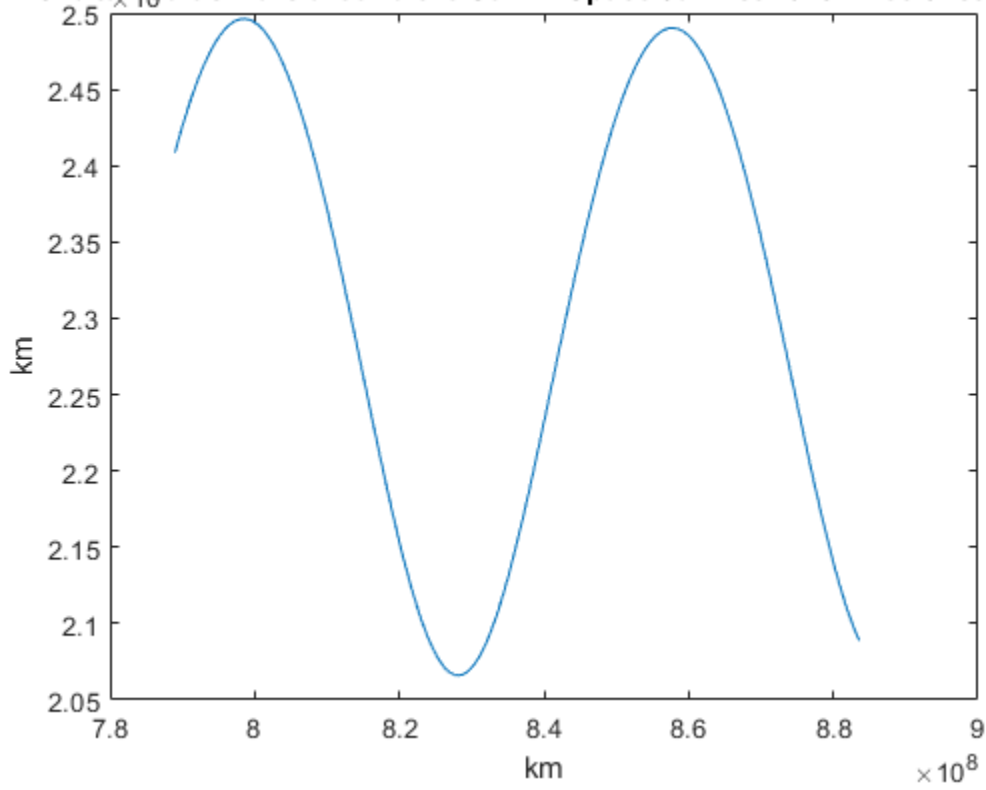
% I notice that the speed propagation for both Moon wrt Earth and Mars wrt
% Sun was essentially inverse to the radius propagation. For example, speed

```

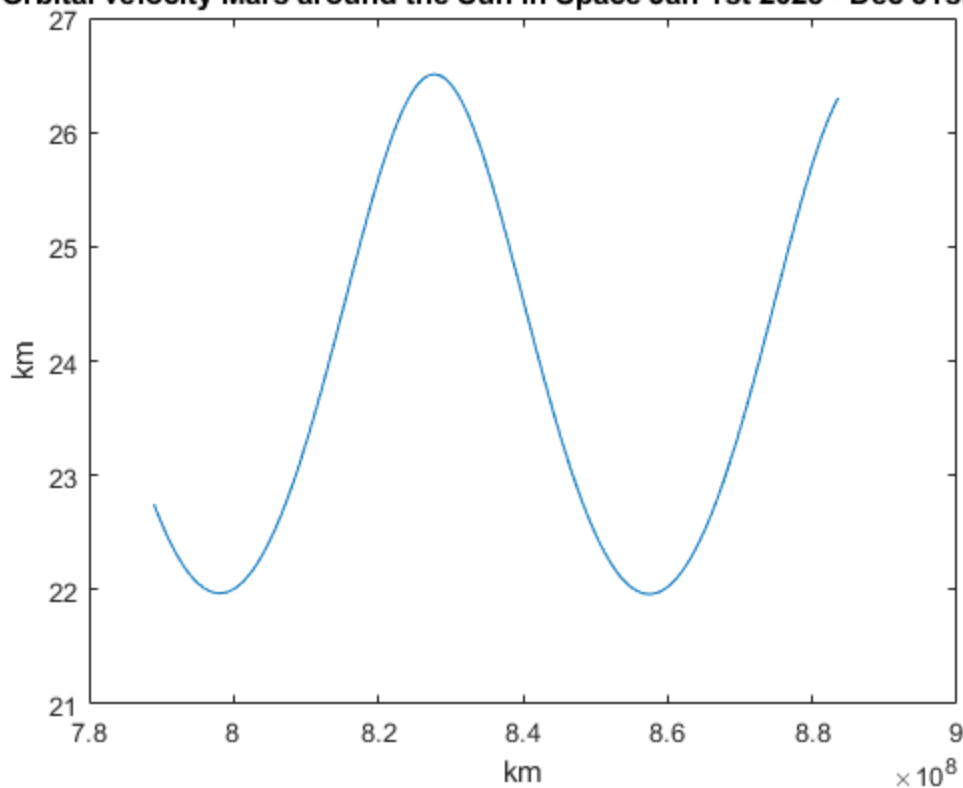
% increases as the radius of the orbit decreases and vice versa. The
 % propogation is sinousodial. However, for the Moon wrt Earth case, the
 % sinousodial pattern is not perfect, and seems to be warped. This could be
 % due to the fact that the sun's gravity still has a powerful impact on the
 % orbit of the Earth and the moon. For the orbits, I noticed that the
 % radius of the orbits for Mercury through Mars is significantly smaller
 % than the orbits from Jupiter to Neptune. Mercury also has a very
 % eccentric orbit.



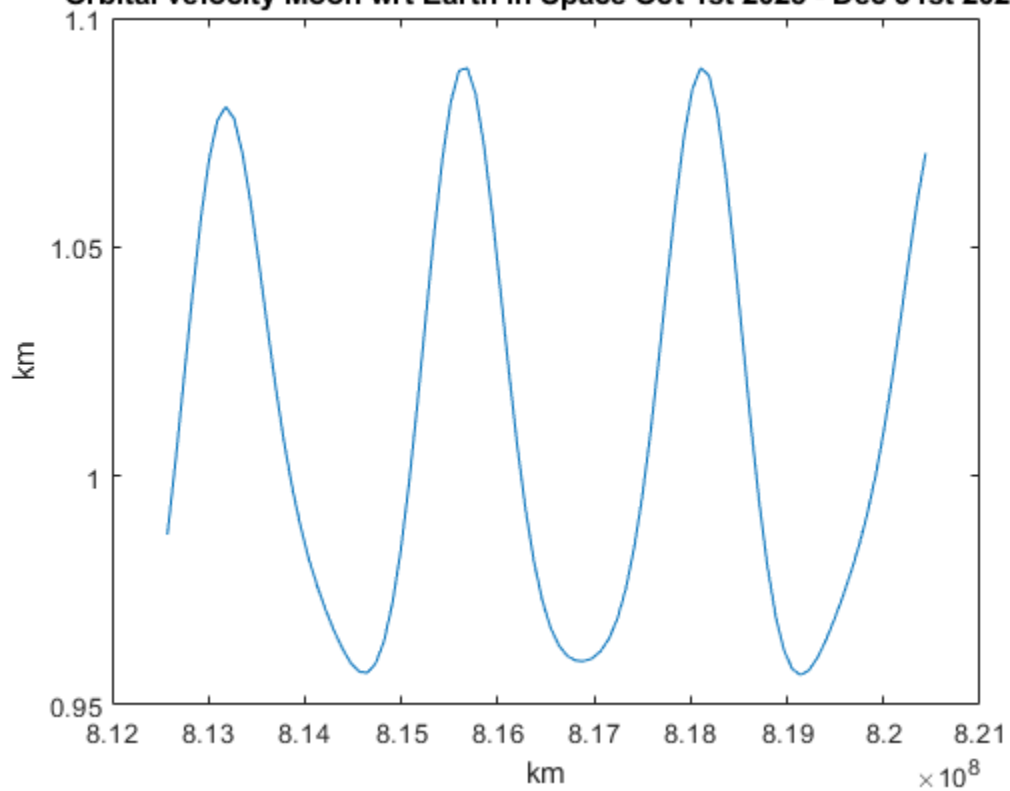
Orbital radius Mars around the Sun in Space Jan 1st 2025 - Dec 31st 2027



Orbital velocity Mars around the Sun in Space Jan 1st 2025 - Dec 31st 2027



Orbital velocity Moon wrt Earth in Space Oct 1st 2025 - Dec 31st 2025



Problem 4

```
% Time and Date

t0_utc = '2025 Jan 01 00:00:00 UTC';
et0 = cspice_str2et(t0_utc);
et_0Y = et0:86400:(et0+(3.15576*10^7)); % make a row vector of times for a
year

% Get mu
mu = 1.32712440018e11;
%GM = cspice_bodvcd(4, 'GM', 10);

% Obtain State Vector from CSpice
MwrtSun = cspice_spkezr('4', et0, 'ECLIPJ2000', 'NONE', '0');
%SwrtSSB= cspice_spkezr('10', et0, 'ECLIPJ2000', 'NONE', '0');

% initial conditions of the trajectory in relation to Mars based on MwrtSun
% state vector

r_0 = MwrtSun(1:3);
v_0 = MwrtSun(4:6);
y_0 = [r_0; v_0];

% Time span (one year for proper comparison to problem 3)
tspan = [0 3600*24*365];

% Use ODE 113 to propagate the 2 body problem equation

two_body_eq = @(t,y) [y(4:6); -mu*y(1:3)/(norm(y(1:3)).^3)]; % Change to fit
your situation

% relative tolerance array
reltol = [1e-4, 1e-6, 1e-8, 1e-10];

% use ode113 to calculate the full orbital trajectories of Mars wrt Sun
% over a year for each relative tolerance type in the reltol array.

opts4 = odeset('RelTol', 1e-4, 'AbsTol', 1e-13);
[t4, Y4] = ode113(two_body_eq, tspan, y_0, opts4);

opts6 = odeset('RelTol', 1e-4, 'AbsTol', 1e-13);
[t6, Y6] = ode113(two_body_eq, tspan, y_0, opts6);

opts8 = odeset('RelTol', 1e-4, 'AbsTol', 1e-13);
[t8, Y8] = ode113(two_body_eq, tspan, y_0, opts8);

opts10 = odeset('RelTol', 1e-4, 'AbsTol', 1e-13);
[t10, Y10] = ode113(two_body_eq, tspan, y_0, opts10);

% Now get the position for the propagated Mars state for each reltol
r4 = Y4(:, 1:3);
r_x4 = Y4(:, 1);
```

```

r_y4 = Y4(:,2);
r_z4 = Y4(:,3);

r6 = Y6(:,1:3);
r_x6 = Y6(:,1);
r_y6 = Y6(:,2);
r_z6 = Y6(:,3);

r8 = Y8(:,1:3);
r_x8 = Y8(:,1);
r_y8 = Y8(:,2);
r_z8 = Y8(:,3);

r10 = Y10(:,1:3);
r_x10 = Y10(:,1);
r_y10 = Y10(:,2);
r_z10 = Y10(:,3);

% Plot full orbital trajectories of MwrtSun from Jan 1st 2025 - Dec 31 2025
% with the different reltols
figure;
hold on;
plot(r_x4,r_y4,'b-','LineWidth', 1.5)
plot(r_x6,r_y6,'r-','LineWidth', 1.5)
plot(r_x8,r_y8,'g-','LineWidth', 1.5)
plot(r_x10,r_y10,'-','LineWidth', 1.5)
xlabel('km');
ylabel('km');
title('Mars Propogated Orbit wrt Sun 2D')
legend('1e-4','1e-6','1e-8','1e-10')
hold off;

% Which RelTol would you use from now on?
%I would use RelTol 1e-10 as it is the most accurate so far. This RelTol
%seems more smooth compared to the rest.

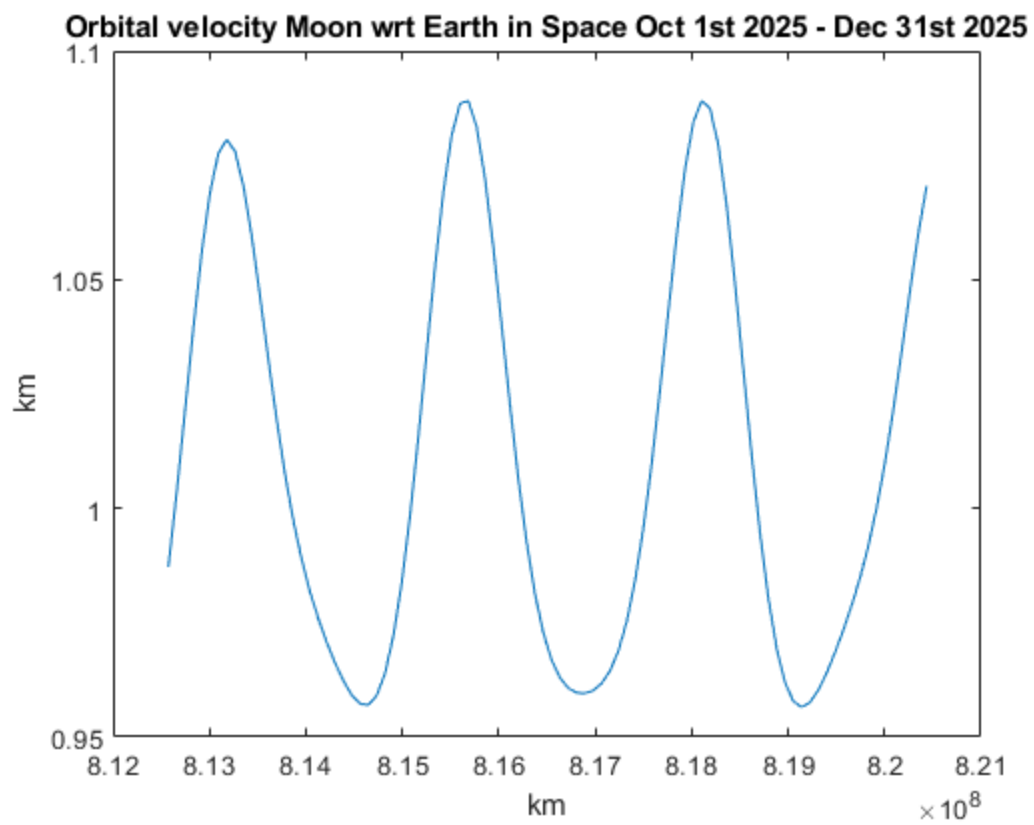
% Compare the propagated states with the ephemerides from SPICE.
% The propogated states line up really closely with the ephemerides from
% SPICE, and is more aligned as the rel tol gets lower. Comparison graph is
% below at RelTol 1e-10.

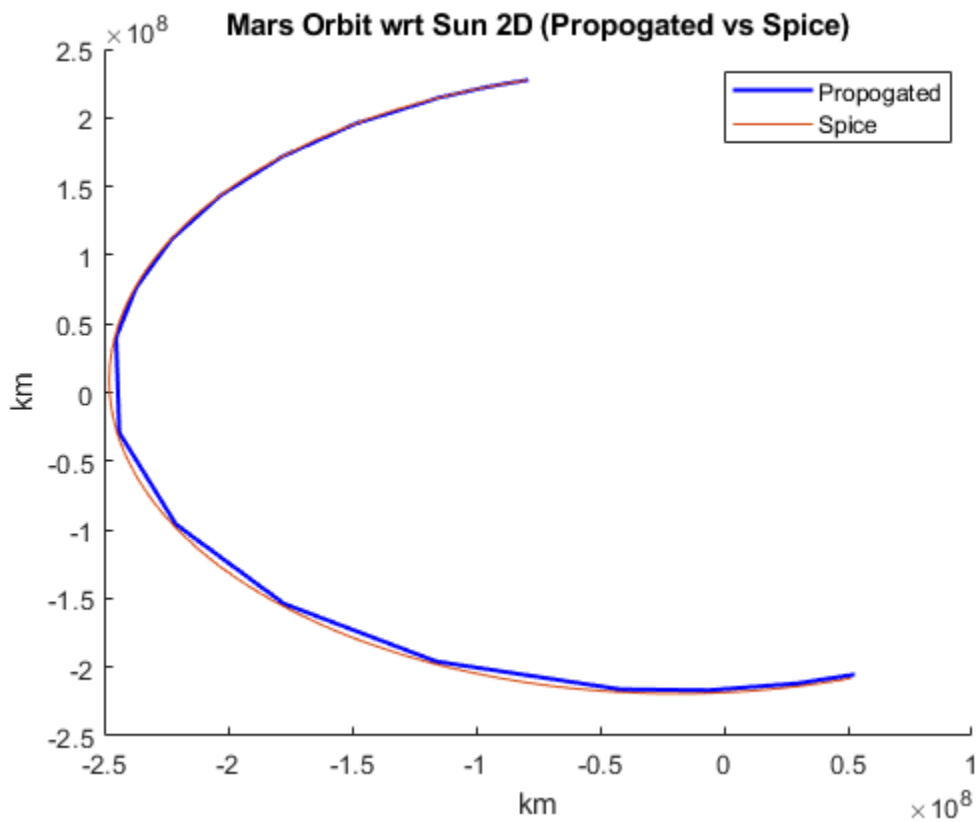
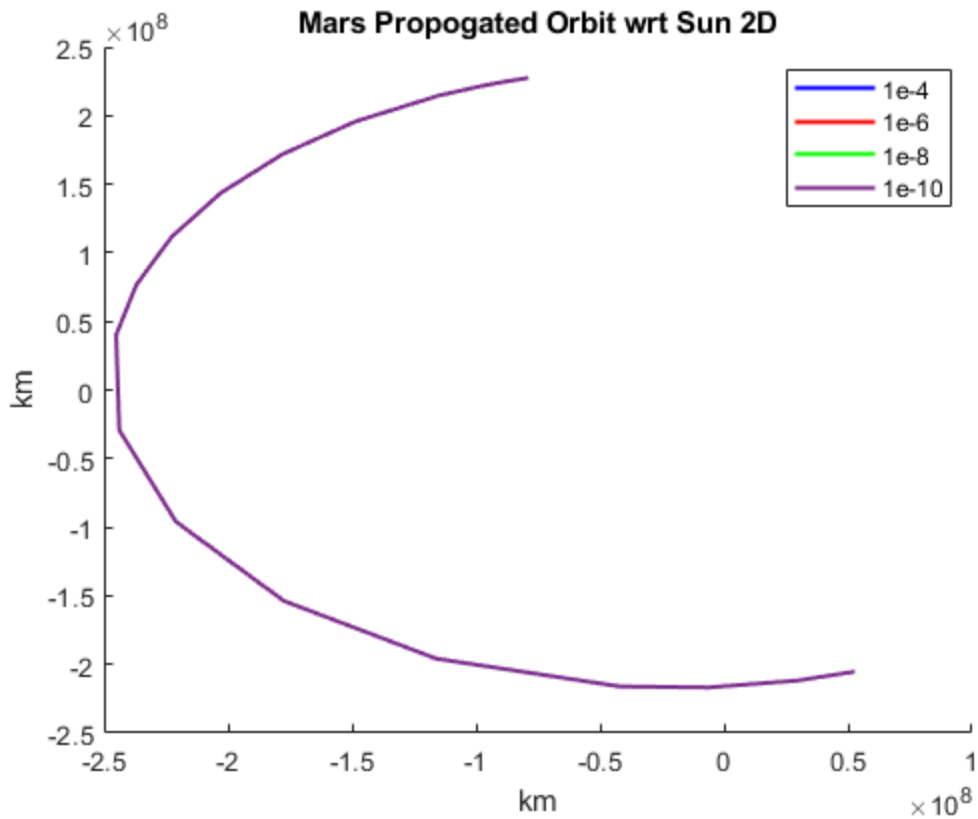
% Getting orbit radius from Spice orbit of Mars throughout one year
pos = zeros(3, 8);
for k = 1:numel(et_Y)
    pos(:, k) = cspice_spkpos('4',et_Y(k), 'ECLIPJ2000', 'NONE', '0');
end

figure;
hold on;
plot(r_x10,r_y10,'b-','LineWidth', 1.5)
plot (pos(1,:), pos(2,:), '-');
xlabel('km');
ylabel('km');

```

```
title('Mars Orbit wrt Sun 2D (Propogated vs Spice)');  
legend('Propogated', 'Spice');
```





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